



**IEEE
ICHI 2026**

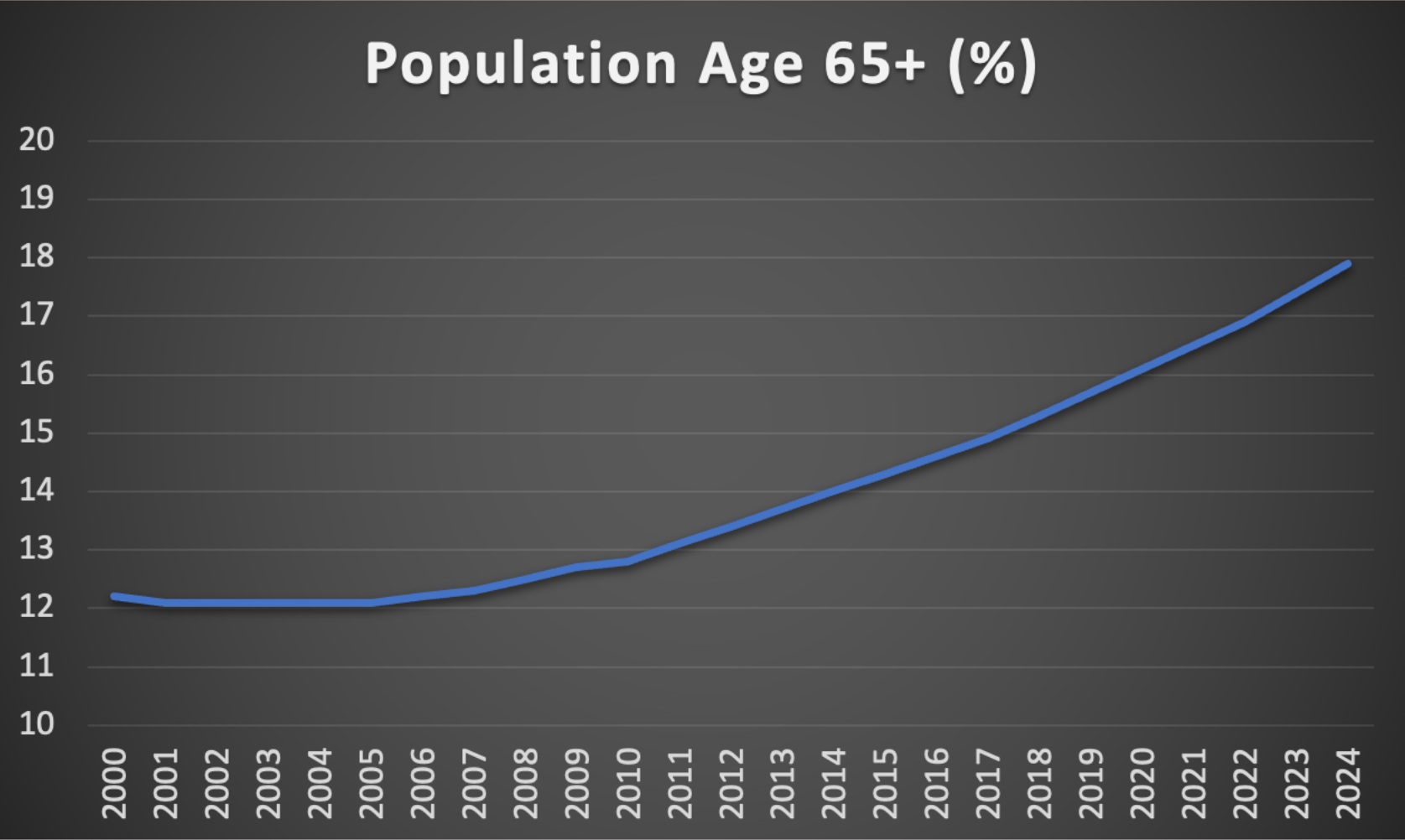
June 1–3, 2026
Minneapolis, MN

Young Scholar Symposium

Emotion-Preserving Music Recommendation from Autobiographical Narratives for Older Adults

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THE AGING POPULATION



61M

THE PROBLEM

>25%

FACE ISOLATION



COGNITIVE DECLINE



MUSIC UNLOCKS MEMORY

- More than a quarter of older adults face social isolation — linked to cognitive decline and depression.
- Research shows music can trigger vivid autobiographical memories, even in people experiencing cognitive decline.
- But no existing system understands someone's life story and recommends emotionally matched music.
- **The goal is memory preservation and emotional validation — not mood correction.**

That's the gap MelodyMemo aims to fill.

RESEARCH QUESTIONS & HYPOTHESES

HYPOTHESIS 1

Can a transformer model reliably extract emotions from personal stories?

Predicted: $\geq 75\%$ accuracy

- DistilRoBERTa transformer classifier
- 7 emotion categories
- 50 autobiographical stories
- Cross-validated with GPT-4o & Gemini

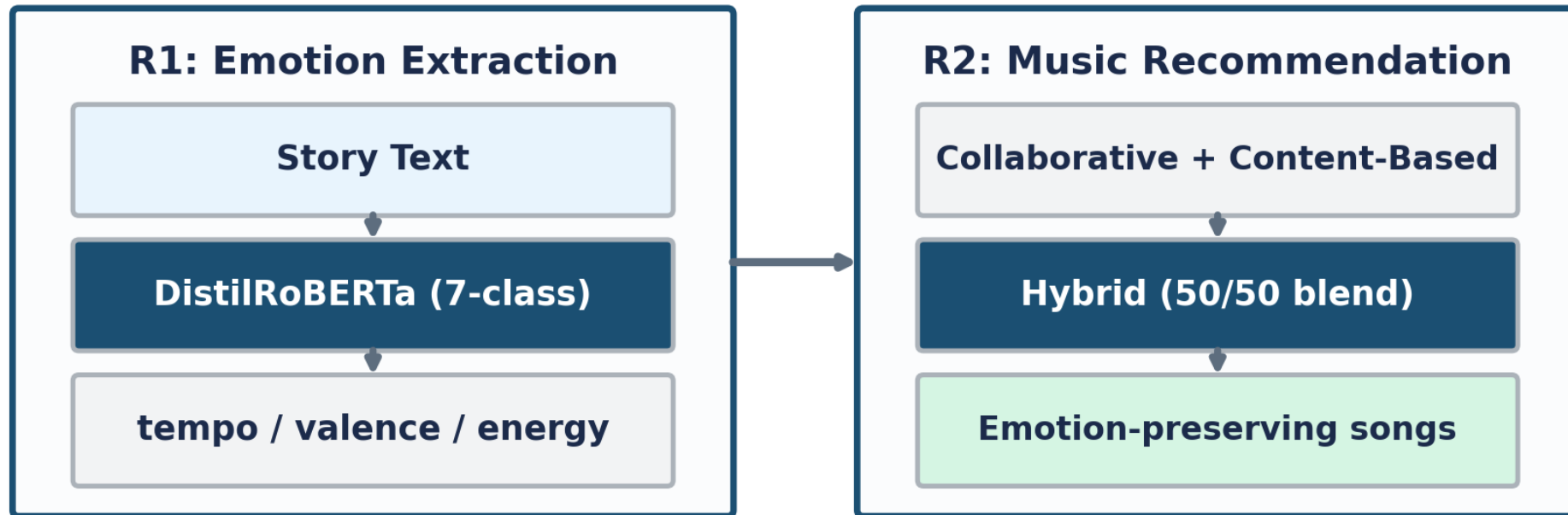
HYPOTHESIS 2

Can a hybrid recommendation algorithm outperform individual methods?

Predicted: hybrid beats both

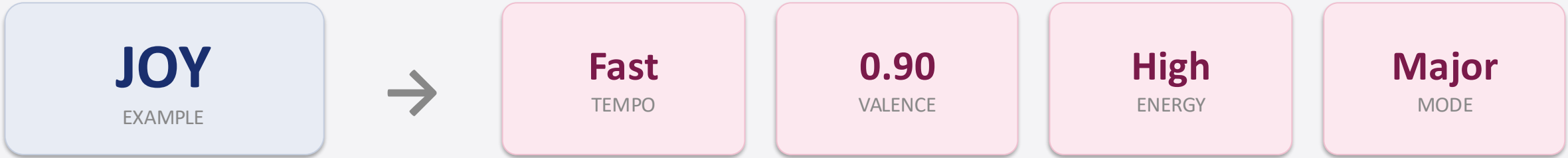
- 3 algorithms: collaborative, content-based, hybrid
- 50 songs from 1939–1975 era
- 25 simulated senior user profiles
- Hybrid = 50% collaborative + 50% content-based

HOW IT WORKS



Story Input → Emotion Detection (R1) → Hybrid Recommendation (R2) → Playlist

EMOTION-TO-MUSIC MAPPING



Complete Emotion → Musical Feature Mapping

Emotion	Tempo	Valence	Energy	Mode
Joy	130 BPM	0.90	0.90	Major
Sadness	70 BPM	0.25	0.30	Minor
Anger	130 BPM	0.30	0.90	Minor
Fear	100 BPM	0.30	0.60	Minor
Love	100 BPM	0.85	0.60	Major
Surprise	130 BPM	0.70	0.80	Major
Neutral	100 BPM	0.50	0.60	Major

R1 RESULTS — EMOTION CLASSIFICATION

80%

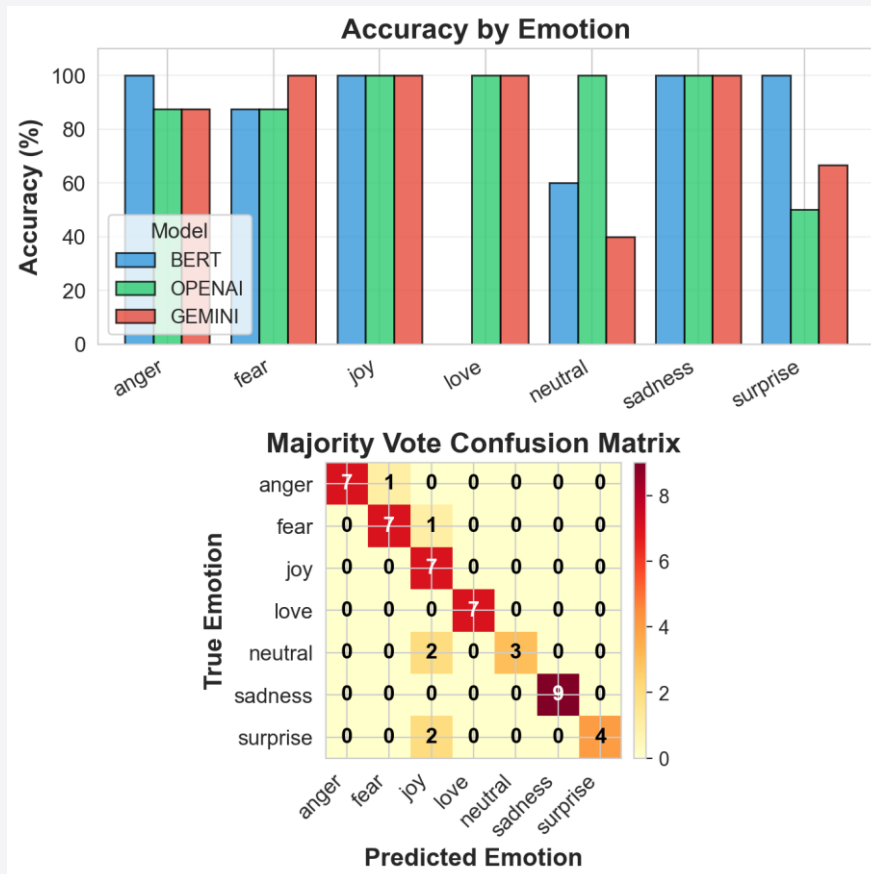
DistilRoBERTa

90%

GPT-4o

88%

GEMINI



✓ H1 Confirmed ($\geq 75\%$)

- 5 of 7 emotions: 87–100%
- Cross-validated with GPT-4o and Gemini
- Love harder due to overlapping vocabulary
- Accurate emotion detection is the foundation for sophisticated recommendations (e.g., recommending uplifting music for sad emotions when appropriate)

R2 RESULTS — RECOMMENDATION

Algorithm	Score	p-value
Random	0.503	—
Collaborative	0.607	< 0.001
Content-Based	0.893	< 0.001
Hybrid (50/50)	0.941	< 0.001

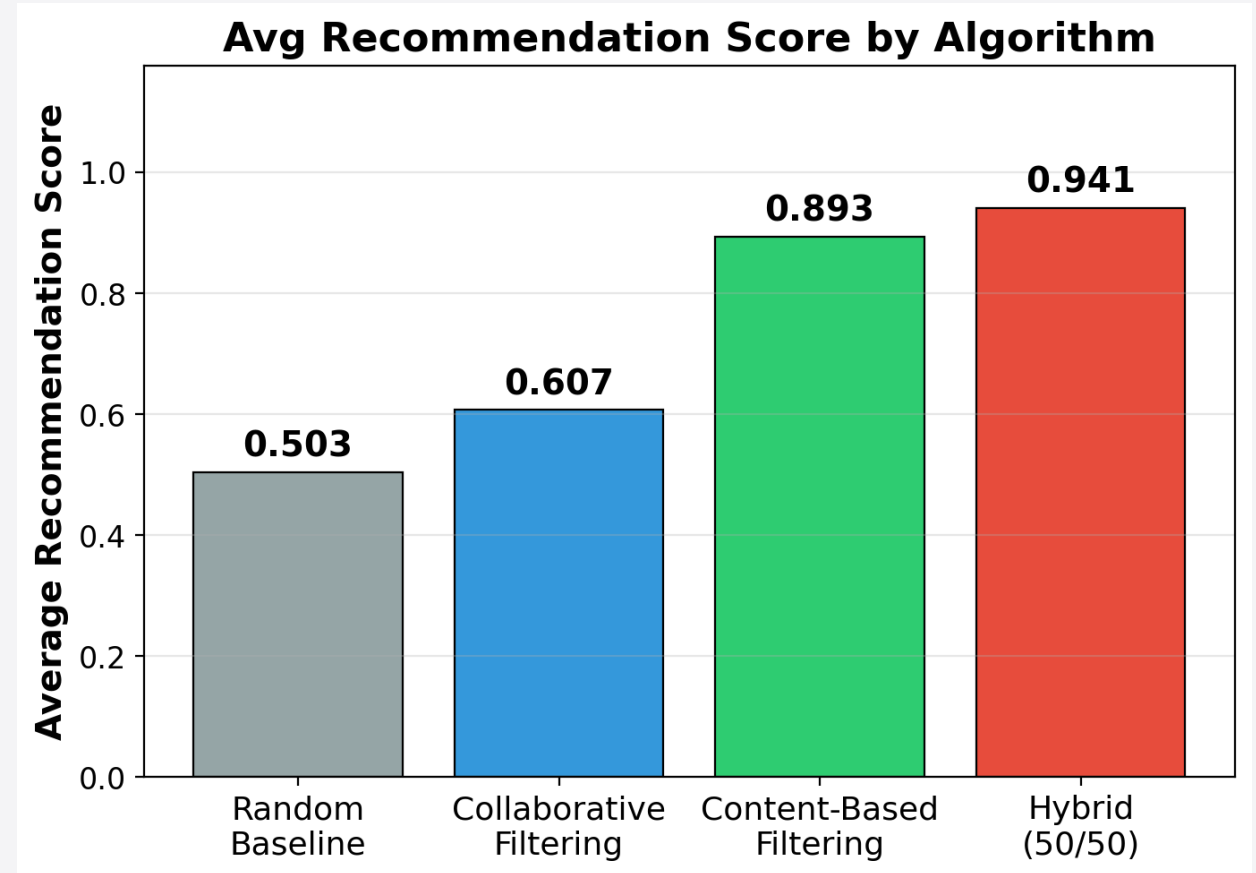
✓ H2 Confirmed — Hybrid outperforms all

Content-based → best emotion alignment

Collaborative → best user ratings

Hybrid captures both strengths

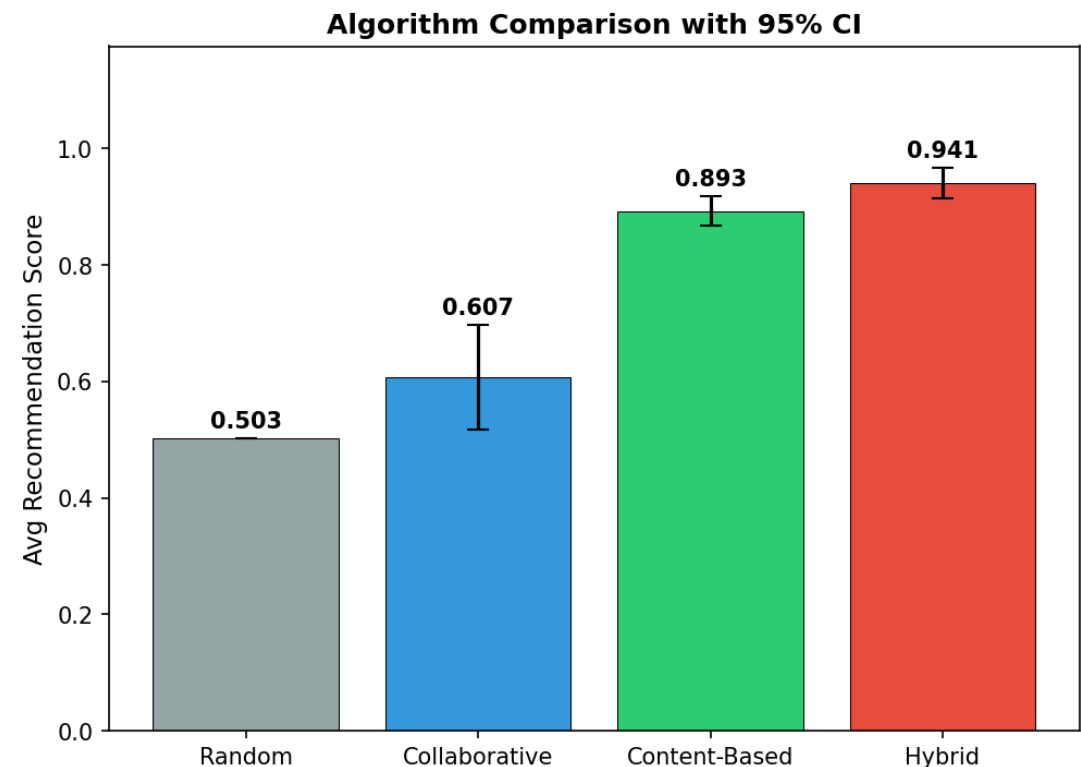
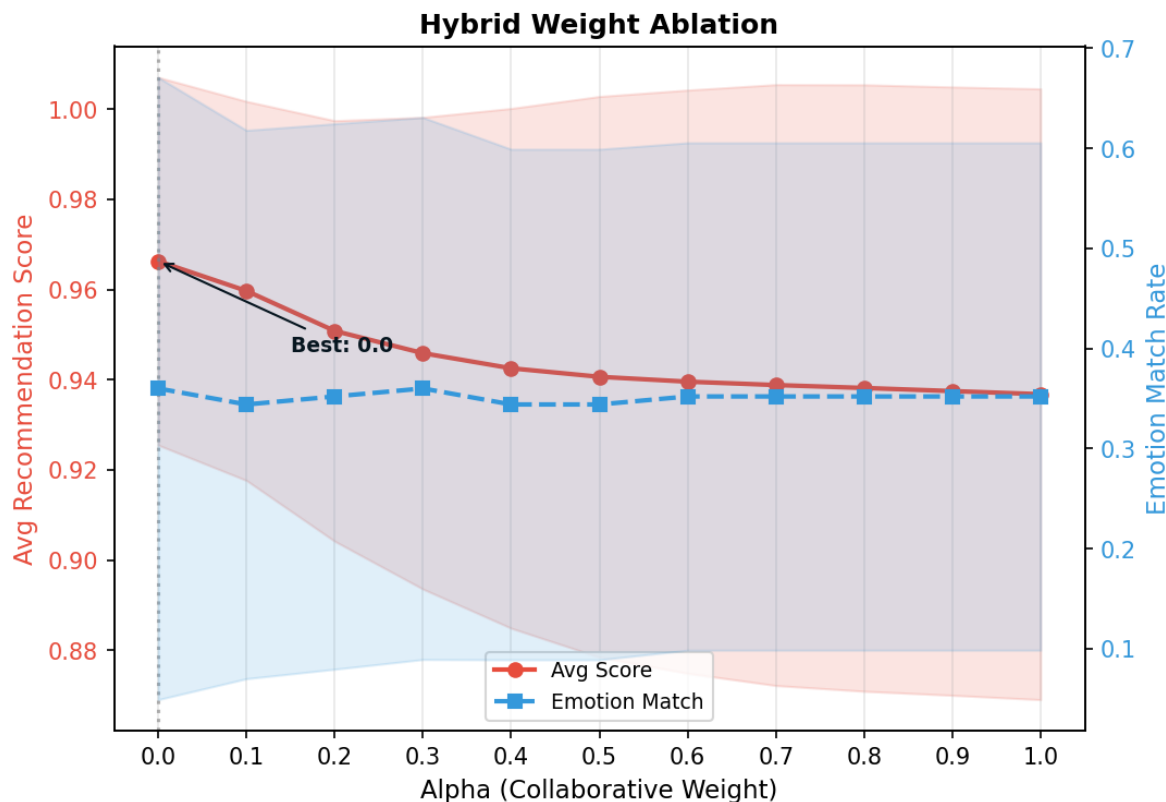
- Understanding emotion enables flexible strategies — match sadness for memory recall, or cross-map to love/joy for mood support



STATISTICAL SIGNIFICANCE

All pairwise comparisons: $p < 0.001$ | The hybrid consistently outperforms the other methods.

R2 Statistical Analysis



NOVEL CONTRIBUTIONS

CONTRIBUTION 1

Transformer NLP for Autobiographical Memory

Applied transformer-based emotion classification to personal narratives written in older adults' voice — a domain not previously explored with this approach.

CONTRIBUTION 2

Systematic Emotion-to Music Feature Mapping

Created a principled mapping from seven emotion categories to musical features (tempo, valence, energy, mode) grounded in music psychology research.

CONTRIBUTION 3

Emotion-Preserving Hybrid Recommendation

Optimized hybrid recommendation to preserve emotional alignment rather than maximizing popularity — achieving 0.941 average score with statistical significance.

CONCLUSIONS & FUTURE WORK

CONCLUSIONS

- ✓ Transformer-based NLP can extract emotions from personal narratives.
- ✓ Emotion-preserving music recommendations are achievable with hybrid algorithms.
- ✓ The hybrid approach (0.941) consistently outperforms individual methods.
- ✓ DistilRoBERTa provides a free, local, private alternative to commercial APIs.

NEXT STEPS

- Pilot study with older adults in music therapy settings
- Fine-tune DistilRoBERTa on older-adult narratives to fix love/neutral errors
- Adaptive personalization using real-time user feedback
- Explore cross-emotion recommendations (e.g., love music for sad stories) as a configurable therapeutic option

Thank you

Questions?

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